

INTERNSHIP SUMMARY REPORT

on the activities carried out during the internship

Host organisation S.A. "Orange Systems"

Faculty of *Computers, Informatics, and Microelectronics*

Study programme *Software Engineering*

Mode of study *full time*

Year of study: 3 **Group:** FAF-243

Internship period: from 01.09.2026 to 26.09.2026

Activities carried out and results obtained:

Activities carried out:

During my internship at Orange Systems, I worked on SecretShare, a web application for sharing confidential information for single retrieval. My contributions focused on the frontend, authentication, and internal network security. I containerised and rewrote the frontend as a Next.js application, replacing the previous static service in docker-compose, and integrated an OAuth2/PKCE authentication flow with Keycloak. To resolve routing issues through Traefik, I switched from subdomain- to path-based routing and removed a conflicting stripPrefix rule so Next.js's basePath resolved properly. I also implemented internal TLS encryption between services. I created a certificate-generation script, configured a dedicated provisioning container, enabled TLS for PostgreSQL and Redis, and enforced CA verification in the API. This required correcting Redis TLS flags and client arguments, injecting TLS contexts into the database migration engine, and adding an automated CI test suite. Additionally, I built a documented turnkey staging environment with a manual deployment pipeline, fixed preview network issues, and implemented the shareable secret-link feature with a burn-after-reading warning.

Results obtained:

SecretShare gained a containerised Next.js frontend with Keycloak authentication and validated reverse-proxy routing. Internal PostgreSQL and Redis traffic is fully TLS-encrypted and verified via internal CA, protected by CI regression testing. The team acquired a documented reference staging environment with manual deployment triggers, and the burn-after-reading link was completed for the MVP. Personally, I developed hands-on experience bridging frontend logic, container orchestration, and infrastructure security.

Difficulties encountered during the internship:

The main challenge was resolving conflicts between Traefik and Next.js routing, specifically identifying that Traefik's stripPrefix rule interfered with Next.js's native basePath handling. The TLS rollout also demanded significant troubleshooting due to library-specific quirks: passing custom keyword arguments to the Redis client and manually injecting TLS contexts into the migration engine. This highlighted how low-level configuration changes ripple across application layers.

Conclusions and recommendations:

The internship bridged theoretical knowledge of authentication and TLS with real-world implementation across a multi-service architecture, showing that a feature is only complete once it reliably survives the reverse proxy, internal networks, and CI/CD pipelines. The TLS rollout was the most valuable experience because its complexity spanned the entire stack: tracing certificates through the API, client drivers, and database migration engines offered clear insight into how infrastructure decisions shape application code. For future projects, I recommend reproducing the proxy and network topology locally early on to prevent elusive routing bugs, alongside logging library-specific configuration quirks immediately so the team avoids duplicate debugging. The company provided a very supportive environment throughout the entire period, and our mentor was genuinely invested in the project, offering invaluable guidance, constructive feedback, and practical tips at every step. Ultimately, this experience strengthened my technical foundation and confirmed my interest in working at the intersection of application development, systems infrastructure, and platform security.

Date of completion: 26/09/2026

Signature of the student: Șoimu Ionuț